

TrES-3b

R-2777

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160716 · created 2026-09-03T23:41:15+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457585.9157 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.057
Transit depth (Rp/Rs)^2	3.24 +/- 2.04 [%]
Orbital Inclination (inc)	86.99 +/- 2.88
Airmass coefficient 1 (a1)	0.91 +/- 0.53
Airmass coefficient 2 (a2)	0.06 +/- 0.28
Scatter in the residuals of the lightcurve fit is	57.29 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	5.75
Transit Duration (day)	0.079 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.18 ± 0.057 vs 0.16309 (deviation 0.16σ)
Duration fit vs published	0.079 d vs 0.0567 d (deviation 1.17σ)
Mid-time O–C	1.7 min
Depth SNR	1.7
Residual scatter	57.29 %

Light curve

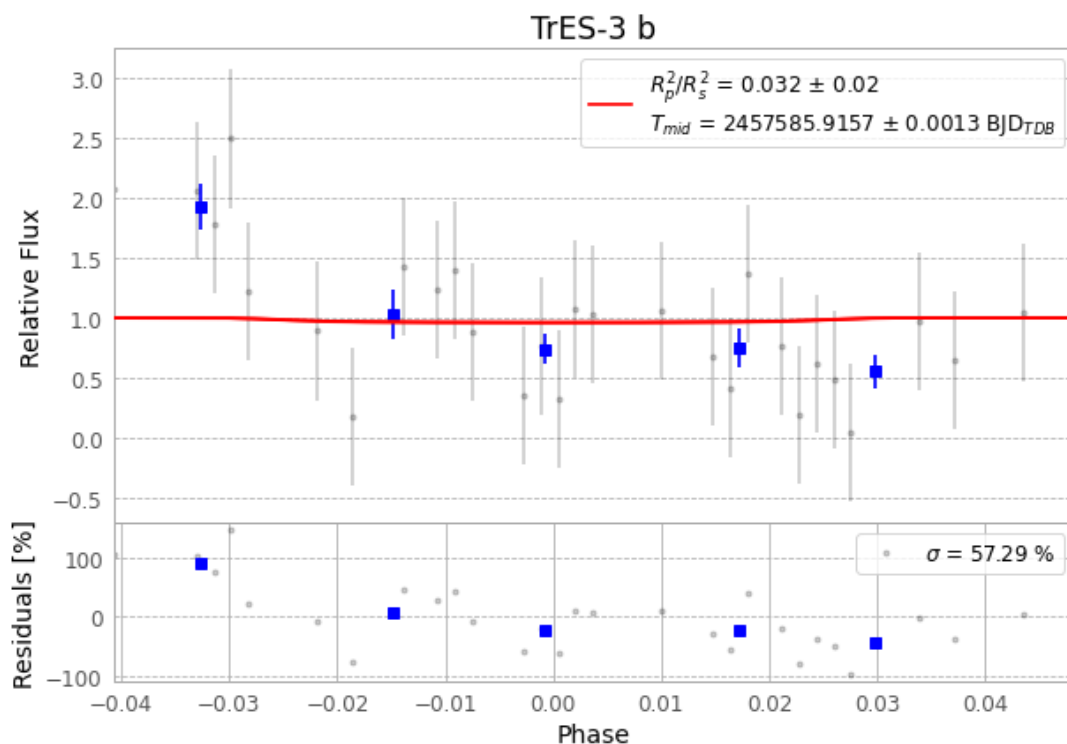


Figure 1 — FinalLightCurve_TrES-3 b_2016-07-16.png (SHA-256 0556cc4916903109...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [271, 180], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 117383e2e03dc81f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

58 FITS frames (38 MB), TRES-3160716083643.FITS → TRES-3160716113012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160716.log (204e1300f2da...), FinalLightCurve_TrES-3 b_2016-07-16.png (0556cc491690...), FinalLightCurve_TrES-3 b_2016-07-16.csv (6e931e8160bf...)

Compute resources

Wall time 182.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 23:41Z.